

The Perceptron Loss Function and the Flexible Perceptron

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Where this fits: Pipeline map step 8 (Model). See the [Course map](#).



- **Builds on:** Stochastic gradient descent ([Note 59](#)); Perceptron trick ([Note 1005](#)).
- **Compare with:** Hinge loss and soft margin ([Note 94](#)).

1. Overview

Key point: A loss function gives every possible line one number: its error. Training a perceptron then means finding the w_1 , w_2 , b with the smallest loss, using gradient descent. Swapping the activation and the loss turns the same perceptron into logistic regression, softmax regression or linear regression.

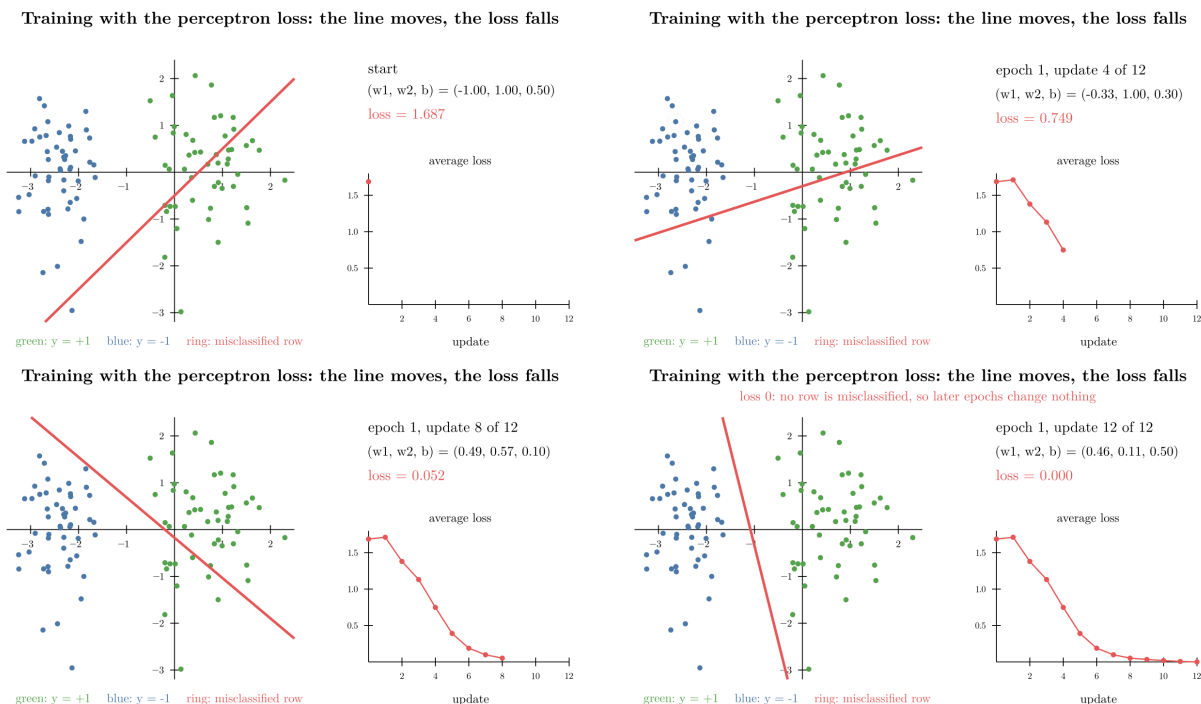


Figure 1: Training with the perceptron loss: the line after 0, 4, 8 and 12 updates, and the average loss

The [perceptron trick Note](#) trained a perceptron by pulling a line towards misclassified points. This Note replaces the trick with a proper method: write down a loss, then minimise it. Figure 1 shows the result: the line moves and the loss falls to 0.

The last section shows why the perceptron is so useful as a building block. The perceptron's design lets us change two parts, the activation and the loss, and get a whole family of models.

2. Prerequisites

- The [perceptron Note](#) and the [perceptron trick Note](#).
- Loss functions and log loss: the [log loss Note](#).
- Gradient descent and partial derivatives: the [gradient descent Note](#) and the [stochastic gradient descent Note](#).

3. What is wrong with the perceptron trick

Key point: The trick cannot tell us how good a line is, and nothing guarantees it settles on a good one.

The perceptron trick usually finds a line that separates the classes. The trick has two weaknesses:

1. **The trick cannot quantify its result.** Two different lines can both separate the training data. Different random orders give different lines (see section 6 of the [perceptron code Note](#)), and the trick has no number that says which one is better.
2. **The trick may not converge.** The trick only moves when it happens to pick a misclassified point, so a long run of correct picks leaves the line where it is. On data that no line can separate, it never stops changing at all (see the [problem with the perceptron Note](#)).

ML algorithms avoid both problems with a [loss function](#): a single number that scores how good the current parameters are.

4. A loss function for a line

Key point: The loss is a function of w_1 , w_2 and b . Every line gets one number; the best line is the one with the smallest number.

A [loss function](#) measures how wrong a model is. For a perceptron with two inputs, the data is fixed, so the loss depends only on the three parameters:

$$L = L(w_1, w_2, b)$$

Changing any of w_1 , w_2 or b moves the line and changes L . Suppose one line gives $L = 25$ and, after moving it, the new line gives $L = 23$. The second line is better. Training means finding the w_1 , w_2 , b where L is smallest; that line is the final model.

Familiar examples: linear regression uses the mean squared error (see the [regression metrics Note](#)), logistic regression the [log loss](#), and the SVM the [hinge loss](#). We can also design our own, as the next section does.

Extra: A loss read as a function of the parameters is also called a cost function (see the [convex and non-convex cost functions Note](#)).

5. Designing a loss for the perceptron

Key point: Counting mistakes treats them all alike. Measuring how far each mistake is from the line is better, and putting the point into the line's equation gives a number proportional to that distance.

5.1 Count the mistakes

Key point: The 0-1 loss counts misclassified points, so a point just over the line costs as much as one far away.

The simplest loss is the number of misclassified points. A line with 7 mistakes is worse than a line with 5. The mistake count is called the **0-1 loss**: each point costs 1 if it is misclassified and 0 if not.

The flaw of the 0-1 loss is that every mistake counts the same. A point just across the line is a small mistake; a point far on the wrong side is a big one. A good loss should weigh mistakes by their size.

Extra: The 0-1 loss is also useless for gradient descent. Moving the line a little usually changes no point's class, so the loss stays flat and its slope is 0 almost everywhere: there is no direction to follow.

5.2 Use the distance

Key point: Add up the perpendicular distances of the misclassified points from the line.

A better loss adds up the perpendicular distance of every misclassified point from the line. With two mistakes at distances $d_1 = 10$ and $d_2 = 13$, the loss is $10 + 13 = 23$. Big mistakes now count more than small ones, much as a speeding fine grows with how far over the limit a driver was, instead of being the same for 1 km/h and 50 km/h.

5.3 Use the value in the line's equation

Key point: Put each misclassified point into $Ax + By + C$ and add up the absolute values. The sum is proportional to the distance and needs only a dot product.

Computing distances takes some work. A simpler number does the same job: put the point into the left-hand side of the line's equation, as in the side test of the [perceptron trick Note](#).

1. **In words:** for every misclassified point, compute $Ax + By + C$, take its absolute value, and add them up.
2. **Formula:**

$$L = \sum_{\text{misclassified}} |Ax_i + By_i + C|$$

3. **Example:** the line $2x + 3y + 4 = 0$ misclassifies the points $(4, 6)$ and $(-2, -2)$:

$$|2 \times 4 + 3 \times 6 + 4| + |2 \times (-2) + 3 \times (-2) + 4| = |30| + |-6| = 30 + 6 = 36$$

The further a point is from the line, the bigger this value. The perceptron loss uses exactly this idea.

Extra: The value is the distance times a fixed number. The distance from a point to the line $Ax + By + C = 0$ is $|Ax + By + C|/\sqrt{A^2 + B^2}$ (see the [equation of a hyperplane Note](#)). Here $\sqrt{2^2 + 3^2} = 3.61$, so the distances are $30/3.61 = 8.32$ and $6/3.61 = 1.66$. Dividing every term by the same 3.61 does not change which line is best.

6. The perceptron loss

Key point: With labels -1 and $+1$, the loss of one point is $\max(0, -y f(x))$: zero when the point is on its correct side, and $|f(x)|$ when it is not.

6.1 The formula

Key point: $L = (1/n) \sum \max(0, -y_i f(x_i))$, where $f(x) = w_1x_1 + w_2x_2 + b$.

scikit-learn's documentation for `SGDClassifier` gives the loss it uses for the perceptron (scikit-learn User Guide, SGD). For this loss, the two classes are labelled $y = +1$ and $y = -1$, not 1 and 0.

1. **In words:** for each **observation** (one record, one row of the data table), multiply its label by its value $f(x)$ in the line's equation and flip the sign; keep it if it is positive, otherwise use 0; then average over all observations.
2. **Formula:** with $f(x_i) = w_1x_{i1} + w_2x_{i2} + b$ (observation i , **features** 1 and 2, where a feature is an input variable, one column of the data table) and n observations,

$$L(w_1, w_2, b) = \frac{1}{n} \sum_{i=1}^n \max(0, -y_i f(x_i))$$

3. **Example:** for the line $2x + 3y + 4 = 0$ and the two misclassified points of Section 5.3, $(4, 6)$ with $y = -1$ and $(-2, -2)$ with $y = +1$:

$$\max(0, -(-1)(30)) + \max(0, -(1)(-6)) = 30 + 6 = 36,$$

and dividing by $n = 2$ gives $L = 18$.

$f(x_i)$ is the perceptron's z for observation i . The quantity $\max(0, s)$ is just " s if s is positive, otherwise 0".

Extra: The full formula in scikit-learn also adds a regularisation term $\alpha R(w)$, as in [Ridge regression](#). We leave it out here (no regularisation).

6.2 What it does in each case

Key point: Correctly classified points add 0; each misclassified point adds $|f(x)|$, which grows with its distance from the line.

A point can relate to the line in four ways:

True label y	Side of the line, sign of $f(x)$	$-yf(x)$	Loss of the point
+1	positive	negative	0 (correct)
-1	negative	negative	0 (correct)
+1	negative	positive	$ f(x) $ (mistake)
-1	positive	positive	$ f(x) $ (mistake)

So the formula is the “value in the line’s equation” loss of Section 5.3, written in one line with no if-statements. The label y only has size 1, so it fixes the sign and the size comes from $f(x)$.

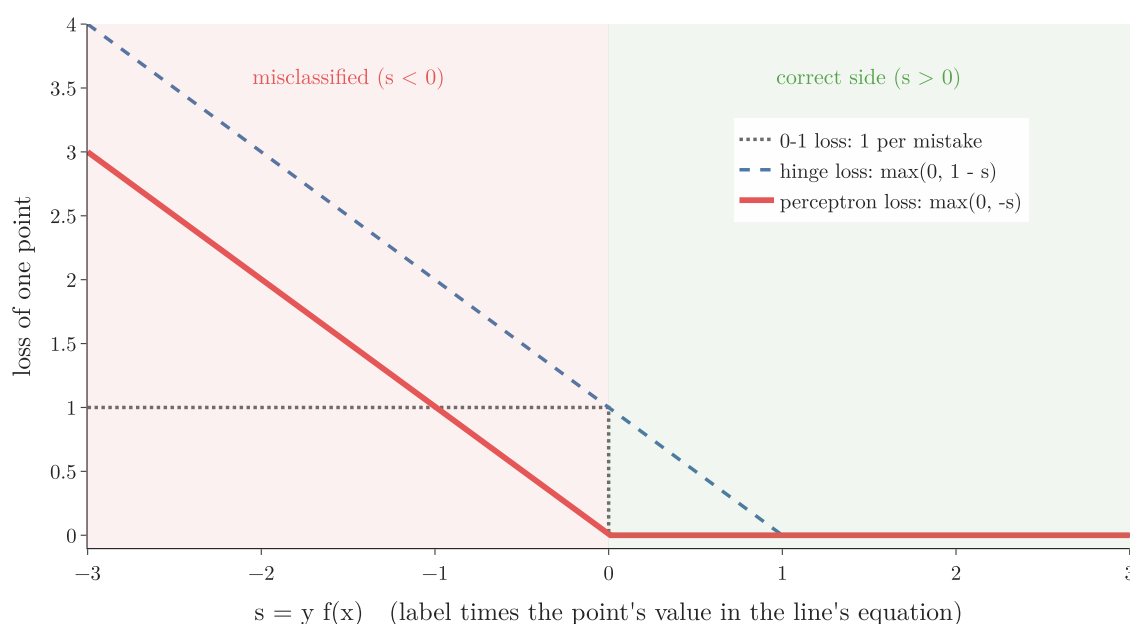


Figure 2: The loss of one point against $s = y f(x)$

Figure 2 plots the loss of one point against $s = y f(x)$. The perceptron loss (red) is 0 for every point on its correct side and rises steadily the further a point is on the wrong side.

Extra: The SVM’s **hinge loss** is $\max(0, 1 - s)$ (blue, dashed): the same shape shifted right by 1. A point must be correct by a margin ($s \geq 1$) before it costs nothing. The perceptron loss accepts any correct point, however close to the line. Once every point is

on its correct side, every observation's gradient is 0 (section 7.1), so the updates stop at the first separating line reached, even one that passes very close to the points of one class.

6.3 Minimising it

Key point: Training is argmin over w_1, w_2, b of L , solved with gradient descent.

Only w_1, w_2 and b can change; the inputs x_{i1}, x_{i2} and labels y_i are fixed by the data. So training is

$$w_1, w_2, b = \arg \min_{w_1, w_2, b} \frac{1}{n} \sum_{i=1}^n \max(0, -y_i f(x_i))$$

argmin means “the values of the variables that make the expression smallest”. We find them with gradient descent: start with any values, then repeat for a number of epochs

$$w_1 \leftarrow w_1 - \eta \frac{\partial L}{\partial w_1}, \quad w_2 \leftarrow w_2 - \eta \frac{\partial L}{\partial w_2}, \quad b \leftarrow b - \eta \frac{\partial L}{\partial b}$$

with a learning rate η such as 0.1 (see the [gradient descent Note](#)).

7. The gradient of the perceptron loss

Key point: For a misclassified observation, $\partial L / \partial w_1 = -y x_1$, $\partial L / \partial w_2 = -y x_2$ and $\partial L / \partial b = -y$; for a correct observation, all three are 0.

7.1 The derivatives

Key point: The chain rule splits the derivative into $\partial L / \partial f$, which is 0 or $-y$, times $\partial f / \partial w_1$, which is x_1 .

Take the loss of one observation, $L_i = \max(0, -y_i f(x_i))$. By the [chain rule](#),

$$\frac{\partial L_i}{\partial w_1} = \frac{\partial L_i}{\partial f} \cdot \frac{\partial f}{\partial w_1}$$

- $\partial f / \partial w_1 = x_{i1}$, because $f = w_1 x_{i1} + w_2 x_{i2} + b$.
- $\partial L_i / \partial f$ depends on which part of the max is active: 0 when $y_i f(x_i) \geq 0$ (the loss is the flat 0), and $-y_i$ when $y_i f(x_i) < 0$ (the loss is $-y_i f$).

Putting them together:

1. **In words:** a correct observation has no slope; a misclassified observation has slope $-y$ times the input (and $-y$ for the bias, whose input is 1).

2. Formula:

$$\frac{\partial L_i}{\partial w_1} = \begin{cases} 0 & y_i f(x_i) \geq 0 \\ -y_i x_{i1} & y_i f(x_i) < 0 \end{cases} \quad \frac{\partial L_i}{\partial w_2} = \begin{cases} 0 & y_i f(x_i) \geq 0 \\ -y_i x_{i2} & y_i f(x_i) < 0 \end{cases} \quad \frac{\partial L_i}{\partial b} = \begin{cases} 0 & y_i f(x_i) \geq 0 \\ -y_i & y_i f(x_i) < 0 \end{cases}$$

3. **Example:** line $2x_1 + 3x_2 + 4 = 0$ and the point $(-2, -2)$ with $y = +1$. Then $f = -6$, so $yf = -6 < 0$: misclassified, loss 6. The slopes are $-1 \times (-2) = 2$, 2 and -1 . One step with $\eta = 0.1$:

$$w_1 = 2 - 0.1 \times 2 = 1.8, \quad w_2 = 3 - 0.1 \times 2 = 2.8, \quad b = 4 - 0.1 \times (-1) = 4.1$$

The point's value becomes $1.8(-2) + 2.8(-2) + 4.1 = -5.1$: its loss drops from 6 to 5.1.

Extra: At exactly $yf = 0$ the max has a corner and no true derivative. Any slope between the two one-sided slopes, 0 and $-y x_1$, is a valid **subgradient** there: the slope of a line that touches the corner and stays below the loss. The code takes 0, because it updates only when $yf < 0$.

7.2 The update rule

Key point: For each misclassified observation: $w \leftarrow w + \eta y x$. The update is the perceptron trick's update, now derived from a loss.

Subtracting η times the slope gives, for a misclassified observation,

$$w_1 \leftarrow w_1 + \eta y_i x_{i1}, \quad w_2 \leftarrow w_2 + \eta y_i x_{i2}, \quad b \leftarrow b + \eta y_i$$

and no change for a correct observation. Updating after every observation, instead of averaging over all observations first, is [stochastic gradient descent](#).

Python: Training with the perceptron loss.

```
def perceptron(X, y, lr=0.1, epochs=1000):
```

```
    # y must hold -1 and +1
```

```
    w1, w2, b = -1.0, 1.0, 0.5
```

```
    for _ in range(epochs):
```

```
        for i in range(X.shape[0]):
```

```
            z = w1 * X[i, 0] + w2 * X[i, 1] + b
```

```
            if y[i] * z < 0: # misclassified
```

```
                w1 += lr * y[i] * X[i, 0]
```

```
                w2 += lr * y[i] * X[i, 1]
```

```
                b += lr * y[i]
```

```
    return w1, w2, b
```

```
y_pm = np.where(y == 1, 1, -1) # 0/1 labels -> -1/+1
```

```
np.where(condition, a, b) takes a where the condition holds and b elsewhere.
```

Figure 1 runs this on 100 points made with `make_classification` (`class_sep=15`, `random_state=41`), starting from the poor line $-x_1 + x_2 + 0.5 = 0$. In the first epoch, 12 observations are misclassified when visited; after those 12 updates the average loss is 0, and later epochs change nothing.

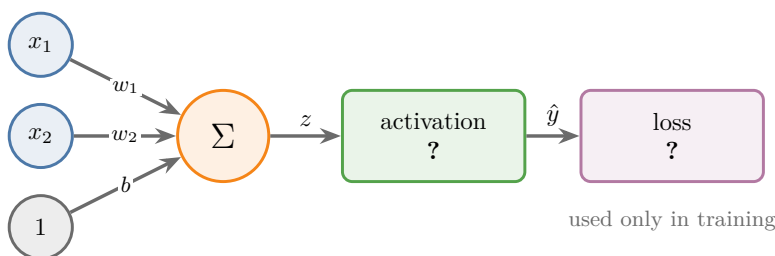
The loss is the average over all observations, while each update looks at one observation. So a single update can raise the total loss: here the first update moves it from 1.687 to 1.712. The trend is still steadily down.

Extra: With labels ± 1 , $\eta y x$ is exactly the perceptron trick's $\eta(y - \hat{y})x$ up to a factor of 2: a positive point on the wrong side gets $+\eta x$, a negative one $-\eta x$. So the trick was gradient descent on the perceptron loss all along. What the loss adds is a number we can watch go down, and compare between lines.

Extra: The labels must be -1 and $+1$. With `make_classification`'s labels 0 and 1, `y[i] * z` is 0 for every class-0 observation, so the condition `< 0` never holds and those observations never update the line. The model then learns only from class 1. From the start line of Figure 1 it ends with all 50 class-0 points misclassified; from the start $(1, 1, 1)$ it happens to work, because class 0 is already on the correct side of that line.

8. One perceptron, many models

Key point: Keep the weighted sum; choose the activation and the loss. Step + perceptron loss is the perceptron, sigmoid + binary cross-entropy is logistic regression, softmax + categorical cross-entropy is softmax regression, and no activation + MSE is linear regression.



Activation	Loss	The perceptron becomes	Output
step	perceptron loss	a perceptron	class +1 or -1
sigmoid	binary cross-entropy	logistic regression	probability (2 classes)
softmax	categorical cross-entropy	softmax regression	probabilities (k classes)
linear (none)	mean squared error	linear regression	any number

Figure 3: The perceptron with a swappable activation and loss

The perceptron’s design has two free slots (Figure 3): the activation function, which shapes the output, and the loss function, used in training. The weighted sum and the training method (gradient descent) stay the same.

- **Step + perceptron loss:** the perceptron of this Note. Output: a class, +1 or -1.
- **Sigmoid + binary cross-entropy:** output a probability between 0 and 1 for two classes. Sigmoid with binary cross-entropy is logistic regression. The loss is

$$L = -\frac{1}{n} \sum_{i=1}^n [y_i \log \hat{y}_i + (1 - y_i) \log(1 - \hat{y}_i)]$$

- **Softmax + categorical cross-entropy:** one output per class, probabilities that add to 1. Softmax with categorical cross-entropy is softmax regression, for more than two classes.
- **Linear (no activation) + mean squared error:** the output is z itself, any number. No activation with mean squared error is linear regression, with loss $\frac{1}{n} \sum (y_i - \hat{y}_i)^2$ (see the [simple linear regression Note](#)).

So “a perceptron is logistic regression” is only true for one choice of the two slots: sigmoid and binary cross-entropy. The same flexibility carries over to every neuron of a neural network: regression networks end in a linear output, and classification networks in a sigmoid or softmax output.

Python: scikit-learn’s `SGDClassifier` trains the same linear model with any of these losses.

```
from sklearn.linear_model import SGDClassifier
```

```
sgd = SGDClassifier(loss="perceptron", penalty=None,  
                    learning_rate="constant", eta0=0.1, random_state=0)  
sgd.fit(X, y)      # labels 0/1 are fine: it converts them itself  
loss="log_loss" gives logistic regression and loss="hinge" a linear SVM. The  
Perceptron class is this same model with eta0=1 (scikit-learn docs, Perceptron).
```

On the 100 points, `Perceptron` and `SGDClassifier(loss="perceptron", eta0=0.1)` find the same line, one with weights ten times the other: $(2.35, -0.19, 2.0)$ and $(0.235, -0.019, 0.2)$. Multiplying all of w_1 , w_2 , b by one positive number leaves the line $z = 0$ where it is. Both classify all 100 points correctly.

9. Summary

Activation	Loss	Model	Output
step	perceptron loss $\max(0, -yf(x))$	perceptron	class ± 1
sigmoid	binary cross-entropy	logistic regression	probability, 2 classes
softmax	categorical cross-entropy	softmax regression	probabilities, k classes
linear (none)	mean squared error	linear regression	any number

- The perceptron trick cannot score a line; a loss function can.
- Counting mistakes (0-1 loss) treats all mistakes alike; the value $|f(x)|$ grows with the distance from the line.
- Perceptron loss: $L = \frac{1}{n} \sum \max(0, -y_i f(x_i))$ with labels ± 1 ; correct points cost 0.
- The loss's gradient for a misclassified observation is $-yx$ (and $-y$ for b), so SGD updates $w \leftarrow w + \eta yx$: the perceptron trick, derived.
- Changing the activation and the loss turns one perceptron into several classic models.

10. Sources

- scikit-learn User Guide, Stochastic Gradient Descent, Mathematical formulation.
- scikit-learn documentation, `sklearn.linear_model.Perceptron` (equivalent to `SGDClassifier(loss="perceptron", eta0=1, learning_rate="constant", penalty=None)`).

11. Key terms

Term	Meaning
0-1 loss	A loss that counts 1 for every misclassified point and 0 for every correct one
Perceptron loss	$\max(0, -yf(x))$ per point, with labels ± 1 : 0 when correct, $ f(x) $ when not
argmin	The values of the variables that make an expression smallest
Subgradient	A slope used at a corner of a function, where the ordinary derivative does not exist
SGDClassifier	scikit-learn's linear classifier trained with SGD, with a choice of loss (perceptron, log loss, hinge, ...)